

Q<sub>no</sub>: 1Binary to Decimal number.(a)  $(100)_2$  :

$$= 1 \times 2^2 + 0 \times 2^1 + 0 \times 2^0$$

$$= 4 + 0 + 0$$

$$= (4)_{10}$$

(b)  $(11010)_2$  :

$$= 1 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 0 \times 2^0$$

$$= 16 + 8 + 0 + 2 + 0$$

$$= 24 + 2$$

$$= (26)_{10}$$

(c)  $(10110010)_2$  :

$$= 1 \times 2^7 + 0 \times 2^6 + 1 \times 2^5 + 1 \times 2^4 + 0 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 0 \times 2^0$$

$$= 128 + 0 + 32 + 16 + 0 + 0 + 2 + 0$$

$$= 128 + 50$$

$$= (178)_{10}$$

(d)  $(1.001)_2$  :

$$= 1 \times 2^0 + 0 \times 2^{-1} + 0 \times 2^{-2} + 1 \times 2^{-3}$$

$$= 1 + 0 + 0 + \frac{1}{8}$$

$$= 1 + 0.125$$

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$$= (1.125)_{10}$$

$$(e) \quad (110100.010011)_2 :$$

$$= 1 \times 2^5 + 1 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 0 \times 2^0 + 0 \times 2^{-1} + 1 \times 2^{-2} + 0 \times 2^{-3} \\ + 0 \times 2^{-4} + 1 \times 2^{-5} + 1 \times 2^{-6}$$

$$= 32 + 16 + 0 + 4 + 0 + 0 + 0 + \frac{1}{4} + 0 + 0 + \frac{1}{32} + \frac{1}{64}$$

$$= 32 + 20 + 0.25 + 0.03125 + 0.0156$$

$$= (59.2969)_{10}$$

$$(f) \quad (11010.10110)_2 :$$

$$= 1 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 0 \times 2^0 + 1 \times 2^{-1} + 0 \times 2^{-2} + 1 \times 2^{-3} \\ + 1 \times 2^{-4} + 0 \times 2^{-5}$$

$$= 16 + 8 + 0 + 2 + 0 + \frac{1}{2} + 0 + \frac{1}{8} + \frac{1}{32} + 0$$

$$= 26 + 0.5 + 0.125 + 0.03125$$

$$= (26.656)_{10}$$

$$(g) \quad (1000001.111)_2 :$$

$$= 1 \times 2^6 + 0 \times 2^5 + 0 \times 2^4 + 0 \times 2^3 + 0 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 + \\ 1 \times 2^{-1} + 1 \times 2^{-2} + 1 \times 2^{-3}$$

$$= 64 + 0 + 0 + 0 + 0 + 0 + 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8}$$

$$= 65 + 0.5 + 0.25 + 0.125$$

$$= (65.875)_{10}$$



$$= (1.125)_{10}$$

$$(e) \quad (110100.010011)_2 :$$

$$= 1 \times 2^5 + 1 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 0 \times 2^0 + 0 \times 2^{-1} + 1 \times 2^{-2} + 0 \times 2^{-3} \\ + 0 \times 2^{-4} + 1 \times 2^{-5} + 1 \times 2^{-6}$$

$$= 32 + 16 + 0 + 4 + 0 + 0 + 0 + \frac{1}{4} + 0 + 0 + \frac{1}{32} + \frac{1}{64}$$

$$= 32 + 20 + 0.25 + 0.03125 + 0.0156$$

$$= (59.2969)_{10}$$

$$(f) \quad (11010.10110)_2 :$$

$$= 1 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 0 \times 2^0 + 1 \times 2^{-1} + 0 \times 2^{-2} + 1 \times 2^{-3} \\ + 1 \times 2^{-4} + 0 \times 2^{-5}$$

$$= 16 + 8 + 0 + 2 + 0 + \frac{1}{2} + 0 + \frac{1}{8} + \frac{1}{32} + 0$$

$$= 26 + 0.5 + 0.125 + 0.03125$$

$$= (26.656)_{10}$$

$$(e) \quad (1000001.111)_2 :$$

$$= 1 \times 2^6 + 0 \times 2^5 + 0 \times 2^4 + 0 \times 2^3 + 0 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 + \\ 1 \times 2^{-1} + 1 \times 2^{-2} + 1 \times 2^{-3}$$

$$= 64 + 0 + 0 + 0 + 0 + 0 + 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8}$$

$$= 65 + 0.5 + 0.25 + 0.125$$

$$= (65.875)_{10}$$

Q No: 2 :

Decimal to Binary number:

(a) 16 :

|   |     |
|---|-----|
| 2 | 16  |
| 2 | 8-0 |
|   | 4-0 |
| 2 | 2-0 |
|   | 1-0 |

$$(16)_{10} = (10000)_2$$

(b) 247 :

|   |       |
|---|-------|
| 2 | 247   |
| 2 | 123-1 |
| 2 | 61-1  |
| 2 | 30-1  |
| 2 | 15-0  |
| 2 | 7-1   |
| 2 | 3-1   |
|   | 1-1   |

$$(247)_{10} = (11110111)_2$$

(c) 962.84 :

$$0.84 \times 2 = 1.68$$

↓

1

$$2 \mid 962$$

$$0.68 \times 2 = 1.36$$

1

$$2 \mid 481-0$$

$$0.36 \times 2 = 0.72$$

0

$$2 \mid 240-1$$

$$0.72 \times 2 = 1.44$$

1

$$2 \mid 120-0$$

$$0.44 \times 2 = 0.88$$

0

$$2 \mid 60-0$$

$$0.88 \times 2 = 1.76$$

1

$$2 \mid 30-0$$

$$2 \mid 15-0$$

$$2 \mid 7-1$$

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$$0.76 \times 2 = 1.52 \quad 1$$

$$0.52 \times 2 = 1.04 \quad 1$$

$$(962.84)_{10} = (1111000010.11010111)_2$$

(d) 0.0132 :

$$0.0132 \times 2 = 0.0264 \quad 0$$

$$0.0264 \times 2 = 0.0528 \quad 0$$

$$0.0528 \times 2 = 0.1056 \quad 0$$

$$0.1056 \times 2 = 0.2112 \quad 0$$

$$0.2112 \times 2 = 0.4224 \quad 0$$

$$0.4224 \times 2 = 0.8448 \quad 0$$

$$0.8448 \times 2 = 1.6896 \quad 1$$

$$0.6896 \times 2 = 1.3792 \quad 1$$

$$0.3792 \times 2 = 0.7584 \quad 0$$

$$0.7584 \times 2 = 1.5168 \quad 1$$

$$0.5168 \times 2 = 1.0336 \quad 1$$

$$0.0336 \times 2 = 0.672 \quad 0$$

$$(0.0132)_{10} = (0.000000110110)_2$$

(e) 6.74 :

$$0.74 \times 2 = 1.48 \quad 1$$

$$0.48 \times 2 = 0.96 \quad 0$$

$$0.96 \times 2 = 1.92 \quad 1$$

$$0.92 \times 2 = 1.84 \quad 1$$

$$0.84 \times 2 = 1.68 \quad 1$$

$$0.68 \times 2 = 1.36 \quad 1$$

$$0.36 \times 2 = 0.72 \quad 0$$

$$\begin{array}{r|l} 2 & 6 \\ \hline 2 & 3-0 \\ & 1-1 \end{array}$$

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 $(674)_{10} : (110101110)_2$ 

Q No: - 3

Octal to Decimal number:

a)  $14_8 :$ 

$$= 1 \times 8^1 + 4 \times 8^0$$

$$= 8 + 4$$

$$= (12)_{10}$$

(b)  $236_8 =$ 

$$= 2 \times 8^2 + 3 \times 8^1 + 6 \times 8^0$$

$$= 128 + 24 + 6$$

$$= (158)_{10}$$

(c)  $1432_8 :$ 

$$= 1 \times 8^3 + 4 \times 8^2 + 3 \times 8^1 + 2 \times 8^0$$

$$= 512 + 256 + 24 + 2$$

$$(794)_{10}$$

(d)  $0.43_8 :$ 

$$= 0 \times 8^0 + 4 \times 8^{-1} + 3 \times 8^{-2}$$

$$= 0 + \frac{4}{8} + \frac{3}{64}$$

$$= 0.5 + 0.046875$$

$$= (0.5469)_{10}$$

(e)  $0.254_8 :$ 

$$= 0 \times 8^0 + 2 \times 8^{-1} + 5 \times 8^{-2} + 4 \times 8^{-3}$$

$$= 0 + \frac{2}{8} + \frac{5}{64} + \frac{4}{512}$$

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$$= 0.25 + 0.078125 + 0.0078125$$

$$= (0.3359)_{10}$$

$$(7) \quad 16742.3_8 :$$

$$= 1 \times 8^4 + 6 \times 8^3 + 7 \times 8^2 + 4 \times 8^1 + 2 \times 8^0 + 3 \times 8^{-1}$$

$$= 4096 + 3072 + 448 + 32 + 2 + \frac{3}{8}$$

$$= 7650 + 0.375$$

$$= 7650.375$$

$$(9) \quad 206.104_8 :$$

$$= 2 \times 8^2 + 0 \times 8^1 + 6 \times 8^0 + 1 \times 8^{-1} + 0 \times 8^{-2} + 4 \times 8^{-3}$$

$$= 128 + 0 + 6 + \frac{1}{8} + 0 + \frac{4}{512}$$

$$= 128 + 6 + 0.125 + 0.0078125$$

$$= (134.1328)_{10}$$

Q No 4 :-

Decimal to Octal :

$$(a) \quad 29 :$$

$$\begin{array}{r|l} 8 & 29 \\ & 3-5 \end{array}$$

$$(29)_{10} = (35)_8$$

$$(b) \quad 68 :$$

$$\begin{array}{r|l} 8 & 68 \\ 8 & 8-4 \\ & 1-0 \end{array}$$

$$(68)_{10} = (104)_8$$

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(c) 125 :

$$(125)_{10} = (175)_8$$

$$\begin{array}{r|l} 8 & 125 \\ 8 & 15-5 \\ \hline & 1-7 \end{array}$$

(d) 243.67 :

$$0.67 \times 8 = 5.36$$

5

$$0.36 \times 8 = 2.88$$

2

$$0.88 \times 8 = 7.04$$

7

$$\begin{array}{r|l} 8 & 243 \\ 8 & 30-3 \\ \hline & 3-6 \end{array}$$

$$(243.67)_{10} = (363.527)_8$$

(e) 419.95 :

$$0.95 \times 8 = 7.6$$

7

$$0.6 \times 8 = 4.8$$

4

$$0.8 \times 8 = 6.4$$

6

$$\begin{array}{r|l} 8 & 419 \\ 8 & 52-3 \\ \hline & 6-4 \end{array}$$

$$(419.95)_{10} = (643.746)_8$$

(f) 645.7 :

$$0.07 \times 8 = 0.56$$

0

$$0.56 \times 8 = 4.48$$

4

$$0.48 \times 8 = 3.84$$

3

$$\begin{array}{r|l} 8 & 645 \\ 8 & 80-5 \\ 8 & 10-0 \\ \hline & 1-2 \end{array}$$

$$(645.7)_{10} = (1205.043)_8$$

(g) 39.4475 :-

$$0.4475 \times 8 = 3.58$$

3

$$0.58 \times 8 = 4.64$$

4

$$0.64 \times 8 = 5.12$$

5

$$\begin{array}{r|l} 8 & 39 \\ & 4-7 \end{array}$$

$$(39.4475)_{10} = (47.345)_8$$

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(h)  $49.21875$  :

$$0.21875 \times 8 = 1.75$$

$$0.75 \times 8 = 6.0$$

$$\begin{array}{r|l} 8 & 49 \\ & 6-1 \end{array}$$

$$(49.21875)_{10} = (61.6)_8$$

Q No: 5

Convert Octal number to their Binary :

(a)  $13_8$  :

$$= 1 \times 8^1 + 3 \times 8^0$$

$$= 8 + 3$$

$$= (11)_{10}$$

$$\begin{array}{r|l} 2 & 11 \\ & 5-1 \\ & 2-1 \\ & 1-0 \end{array}$$

$$(13)_8 = (1011)_2$$

(b)  $27_8$  :

$$= 2 \times 8^1 + 7 \times 8^0$$

$$= 16 + 7$$

$$= (23)_{10}$$

$$\begin{array}{r|l} 2 & 23 \\ & 11-1 \\ & 5-1 \\ & 2-1 \\ & 1-0 \end{array}$$

$$(27)_8 = (10111)_2$$

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$$\begin{aligned}
 (c) \quad 65_{10} &:= \\
 &= 6 \times 10^1 + 5 \times 10^0 \\
 &= 40 + 5 \\
 &= (53)_{10}
 \end{aligned}$$

$$(65)_{10} = (110101)_2$$

$$\begin{array}{r|l}
 2 & 53 \\
 \hline
 2 & 26-1 \\
 2 & 13-0 \\
 2 & 6-1 \\
 2 & 3-0 \\
 & 1-1
 \end{array}$$

$$(d) \quad 124.375 :$$

$$= 1 \times 10^2 + 2 \times 10^1 + 4 \times 10^0 + 3 \times 10^{-1} + 7 \times 10^{-2} + 5 \times 10^{-3}$$

$$= 64 + 16 + 4 + 3/10 + 7/100 + 5/1000$$

$$= 84 + 0.375 + 0.109375 + 0.009765625$$

$$= (84.494)_{10}$$

$$0.494 \times 2 = 0.988 \quad 0$$

$$0.988 \times 2 = 1.976 \quad 1$$

$$0.976 \times 2 = 1.952 \quad 1$$

$$0.952 \times 2 = 1.904 \quad 1$$

$$0.904 \times 2 = 1.808 \quad 1$$

$$0.808 \times 2 = 1.616 \quad 1$$

$$0.616 \times 2 = 1.232 \quad 1$$

$$0.232 \times 2 = 0.464 \quad 0$$

$$0.464 \times 2 = 0.928 \quad 0$$

$$0.928 \times 2 = 1.856 \quad 1$$

$$(124.375)_{10} = (1010100.011111001)_2$$

نوٹ: ہر عدد "values" ہو گی انہی بار

E.g:- 84.494

"Multiply" کرنا ہے۔

صرف انہی بار "Multiply" کرنا ہے

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$$(d) \ 10111.101$$

$$= 1 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 + 1 \times 2^{-1} + 0 \times 2^{-2} + 1 \times 2^{-3}$$

$$= 16 + 0 + 4 + 2 + 1 + \frac{1}{2} + 0 + \frac{1}{8}$$

$$= 23 + 0.5 + 0.125$$

$$(23.625)_{10}$$

$$0.625 \times 8 = 5$$

$$\begin{array}{r|l} 8 & 23 \\ \hline & 2-7 \end{array}$$

$$(10111.101)_2 = (27.5)_8$$

$$(e) \ 111010.001$$

$$= 1 \times 2^5 + 1 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 0 \times 2^0 + 0 \times 2^{-1} + 0 \times 2^{-2} + 1 \times 2^{-3}$$

$$= 32 + 16 + 8 + 0 + 2 + 0 + 0 + 0 + \frac{1}{8}$$

$$= 58 + 0.125$$

$$= (58.125)_{10}$$

$$0.125 \times 8 = 1$$

$$\begin{array}{r|l} 8 & 58 \\ \hline & 7-2 \end{array}$$

$$111010.001 = (72.1)_8$$

Q no: 9

Addition of Binary number:

$$\begin{array}{r} a) \quad \begin{array}{ccc} & 1 & 1 \\ + & 1 & 1 \\ \hline 1 & 1 & 0 \end{array} \end{array}$$

$$\begin{array}{r} (b) \quad \begin{array}{cccc} & 0 & 0 & 0 \\ & 1 & 0 & 1 & 1 \\ + & 1 & 1 & 0 & 1 \\ \hline 1 & 1 & 0 & 0 & 0 \end{array} \end{array}$$

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$$\begin{array}{r}
 00000 \\
 (c) \ 101011 \\
 110101 \\
 \hline
 1100000
 \end{array}$$

$$\begin{array}{r}
 000000 \\
 (d) \ 1110101 \\
 1011111 \\
 \hline
 11010100
 \end{array}$$

Q No: 10

Addition of Binary Numbers:

$$\begin{array}{r}
 0001 \\
 (a) \ 1101 \\
 1011 \\
 \hline
 100011 \\
 000110 \\
 \hline
 100110
 \end{array}$$

$$\begin{array}{r}
 00000 \\
 1010110 \\
 111010 \\
 \hline
 100010101 \\
 00010101 \\
 \hline
 100110110
 \end{array}$$

$$\begin{array}{r}
 (c) \ 1111011 \\
 1011111 \\
 \hline
 100110000
 \end{array}$$

$$\begin{array}{r}
 11010101 \\
 11010110 \\
 \hline
 1001100001
 \end{array}$$

Q No: 11

Subtract following binary numbers.

$$\begin{array}{r}
 (a) \ 11 \\
 - 10 \\
 \hline
 01
 \end{array}$$

$$\begin{array}{r}
 (b) \ 111 \\
 - 101 \\
 \hline
 010
 \end{array}$$

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$$\begin{array}{r}
 (c) \quad 1 \quad 0 \quad 1 \quad 0 \quad 1 \quad 1 \\
 - \quad 1 \quad 0 \quad 0 \quad 1 \quad 0 \quad 1 \\
 \hline
 0 \quad 0 \quad 0 \quad 1 \quad 1 \quad 0
 \end{array}$$

$$\begin{array}{r}
 (d) \quad 1 \quad 1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 1 \\
 - \quad 1 \quad 0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \\
 \hline
 0 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0
 \end{array}$$

Q no: 12 :-

Multiply the following.

$$\begin{array}{r}
 (a) \quad 1 \quad 1 \\
 \times 1 \quad 1 \\
 \hline
 1 \quad 1 \\
 1 \quad 1 \quad x \\
 \hline
 1 \quad 0 \quad 0 \quad 1
 \end{array}$$

$$\begin{array}{r}
 (b) \quad 1 \quad 0 \quad 1 \\
 \quad \quad 1 \quad 0 \\
 \quad \quad 0 \quad 0 \quad 0 \\
 \quad \quad 1 \quad 0 \quad 1 \quad x \\
 \hline
 1 \quad 0 \quad 1 \quad 0
 \end{array}$$

$$\begin{array}{r}
 (c) \quad 1 \quad 1 \quad 0 \\
 \times 1 \quad 0 \quad 1 \\
 \hline
 1 \quad 1 \quad 0 \\
 0 \quad 0 \quad 0 \quad x \\
 1 \quad 1 \quad 0 \quad x \quad x \\
 \hline
 1 \quad 1 \quad 1 \quad 1 \quad 0
 \end{array}$$

$$\begin{array}{r}
 (d) \quad 1 \quad 0 \quad 1 \quad 0 \\
 \times 1 \quad 0 \quad 1 \\
 \hline
 1 \quad 1 \quad 1 \quad 0 \\
 0 \quad 0 \quad 0 \quad 0 \quad x \\
 1 \quad 0 \quad 1 \quad 0 \quad x \quad x \\
 \hline
 1 \quad 1 \quad 0 \quad 1 \quad 1 \quad 0
 \end{array}$$

$$(e) \quad 1 \quad 0 \quad 1 \quad 1$$

$$\times 1 \quad 1 \quad 0 \quad 1$$

$$\hline
 1 \quad 0 \quad 1 \quad 1$$

$$0 \quad 0 \quad 0 \quad 0 \quad x$$

$$1 \quad 0 \quad 1 \quad 1 \quad x \quad x$$

$$1 \quad 0 \quad 1 \quad 1 \quad x \quad x$$

$$1 \quad 0 \quad 0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 1$$

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Q. No: - 7

$18 \times 24$  to binary numbers  
 $= 18 \times 24$   
 $= (432)_{10}$

$18 \times 24 = (110110000)_2$

$$\begin{array}{r}
 2 \overline{) 432} \\
 \underline{216} \phantom{0} - 0 \\
 2 \overline{) 216} \\
 \underline{108} \phantom{0} - 0 \\
 2 \overline{) 108} \\
 \underline{54} \phantom{0} - 0 \\
 2 \overline{) 54} \\
 \underline{27} \phantom{0} - 0 \\
 2 \overline{) 27} \\
 \underline{13} \phantom{0} - 1 \\
 2 \overline{) 13} \\
 \underline{6} \phantom{0} - 1 \\
 2 \overline{) 6} \\
 \underline{3} \phantom{0} - 0 \\
 2 \overline{) 3} \\
 \underline{1} \phantom{0} - 1
 \end{array}$$

## Short Questions

Q. No: 1 :

Define Binary number:

Binary number system is the number system of base or radix equal to 2; which means that there are two symbols used to represent number 0 and 1.



Q no: 2 :-

### Octal Number :

The Octal number system is used extensively in digital work because it is easy to convert from octal to binary, vice versa.

The octal system has a base or radix of 8 which means that there are 8 symbols which are used to form octal number.

Q no: 3

### Decimal Number :

The decimal number system is a number of a base or radix equal to 10, which means that there are 10, called Arabic number symbol used to represent

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number: 0, 1, 2, 3, ..., 9

Q No: 4 :-

Binary to Decimal no.

$$\begin{aligned}
 & (10101)_2 \\
 &= 1 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 \\
 &= 16 + 0 + 4 + 0 + 1 \\
 &= (21)_{10}
 \end{aligned}$$

Q No: 5 :-

Binary to Decimal :

$$\begin{aligned}
 & (11111)_2 \\
 &= 1 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 \\
 &= 16 + 8 + 4 + 2 + 1 \\
 &= (29)_{10}
 \end{aligned}$$

Q No: 6 :-

Binary to Decimal

$$\begin{aligned}
 & (110011.11)_2 \\
 &= 1 \times 2^5 + 1 \times 2^4 + 0 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 + 1 \times 2^{-1} + 1 \times 2^{-2} \\
 &= 32 + 16 + 0 + 0 + 2 + 1 + \frac{1}{2} + \frac{1}{4} \\
 &= 51 + 0.5 + 0.25 \\
 &= (51.75)_{10}
 \end{aligned}$$

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Q no: 7

Binary Addition :

$$\begin{array}{r} (1 \ 0 \ 1 \ 1)_2 \\ + (1 \ 1 \ 0)_2 \\ \hline (1 \ 0 \ 0 \ 0 \ 1)_2 \end{array}$$

Q no: 8

Binary Addition :

$$\begin{array}{r} (0 \ 1 \ 0 \ 0 \ 1 \ 1 \ 1)_2 \\ + (1 \ 1 \ 0 \ 1 \ 1 \ 0 \ 1)_2 \\ \hline (1 \ 0 \ 0 \ 1 \ 0 \ 1 \ 0 \ 0)_2 \end{array}$$

Q no: 9

Multiply of Binary Numbers

$$\begin{array}{r} (1 \ 1 \ 1)_2 \\ \times (1 \ 0 \ 1)_2 \\ \hline 1 \ 1 \ 1 \\ 0 \ 0 \ 0 \ x \\ 1 \ 1 \ 1 \ x \ x \\ \hline (1 \ 0 \ 0 \ 0 \ 1 \ 1)_2 \end{array}$$

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Q No: 70

Subtract the BinaryNumber:

$$\begin{array}{r}
 (1 \ 1 \ 0 \ 0)_2 \\
 - (1 \ 0 \ 0 \ 1)_2 \\
 \hline
 (0 \ 0 \ 1 \ 1)_2
 \end{array}$$

Q No: 12

Binary to Octal

$$\begin{aligned}
 & (101101)_2 \\
 &= 1 \times 2^5 + 0 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 \\
 &= 32 + 0 + 8 + 4 + 0 + 1 \\
 &= (45)_{10}
 \end{aligned}$$

$$(101101)_2 = (55)_8$$

$$\begin{array}{r}
 8 \overline{) 45} \\
 \underline{5-5} \\
 0
 \end{array}$$

Q No: 13

Binary to Octal

$$\begin{aligned}
 & (10110111)_2 \\
 &= 1 \times 2^7 + 0 \times 2^6 + 1 \times 2^5 + 1 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 \\
 &= 128 + 0 + 32 + 16 + 0 + 4 + 2 + 1 \\
 &= (183)_{10}
 \end{aligned}$$

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$$(10110111)_2 = (255)_8$$

$$\begin{array}{r} 8 \overline{) 103} \\ 8 \overline{) 11-5} \\ \underline{2-5} \end{array}$$

Q No: 14

Binary to Octal number:

$$(110110.011)_2$$

$$= 1 \times 2^5 + 1 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 0 \times 2^0 + 0 \times 2^{-1} + 1 \times 2^{-2} + 1 \times 2^{-3}$$

$$= 32 + 16 + 0 + 4 + 2 + 0 + 0 + \frac{1}{4} + \frac{1}{8}$$

$$= 54 + 0.25 + 0.125$$

$$= (54.375)_{10}$$

$$0.375 \times 8 = 3.0 \quad \downarrow \quad \begin{array}{r} 8 \overline{) 54} \\ 6-6 \end{array}$$

$$(110110.011)_2 = (66.3)_8$$

Q No: 15

Octal to Binary:

$$(103.45)_8$$

$$= 1 \times 8^2 + 0 \times 8^1 + 3 \times 8^0 + 4 \times 8^{-1} + 5 \times 8^{-2}$$

$$= 64 + 0 + 3 + \frac{4}{8} + \frac{5}{64}$$

$$= 67 + 0.5 + 0.078125$$

$$(67.58)_{10}$$

$$0.58 \times 2 = 1.16 \quad \downarrow \quad \begin{array}{r} 2 \overline{) 67} \\ 2 \overline{) 33-1} \\ 2 \overline{) 16-1} \\ 2 \overline{) 8-0} \\ 1 \overline{) 4-0} \\ 2 \overline{) 2-0} \\ 1 \overline{) 0} \end{array}$$

$$0.16 \times 2 = 0.32 \quad 0$$

$$(103.45)_8 = (1000011.10)_2$$

Teacher Feedback \_\_\_\_\_

Date: .....

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Q No: 16

Octal number to Binary

$$\begin{aligned}
 & (107)_8 \\
 &= 1 \times 8^2 + 0 \times 8^1 + 7 \times 8^0 \\
 &= 64 + 0 + 7 \\
 &= (71)_{10}
 \end{aligned}$$

$$\begin{array}{r|l}
 2 & 71 \\
 \hline
 2 & 35-1 \\
 \hline
 2 & 17-1 \\
 \hline
 2 & 8-1 \\
 \hline
 2 & 4-0 \\
 \hline
 2 & 2-0 \\
 \hline
 & 1-0
 \end{array}$$

$$(107)_8 = (1000111)_2$$

Q.No: 17

Octal equivalent

$$\begin{aligned}
 & (359.325)_{10} \\
 & 0.325 \times 8 = 2.5 \quad 2 \\
 & 0.5 \times 8 = 4.0 \quad 4 \\
 & 0.0 \times 8 = 0.0 \quad 0
 \end{aligned}$$

$$\begin{array}{r|l}
 8 & 359 \\
 \hline
 8 & 44-5 \\
 \hline
 & 5-4
 \end{array}$$

$$(359.325)_{10} = (545.246)_8$$

Marks: ..... Percent ..... % Grade .....



Q No: 18

Decimal to Octal: $(932)_{10}$ 

$$(932)_{10} = (1500)_8$$

|   |       |
|---|-------|
| 8 | 932   |
| 8 | 104-0 |
| 8 | 13-0  |
|   | 15    |

Exercise

Q. No: 1 :-

Find frequency, cumulative frequency, Relative frequency class boundary and tellegam.

81, 86, 82, 76, 92, 89, 87, 82, 88

83, 85, 91, 77, 93, 83, 98, 92, 87

87, 82, 71, 88, 78, 84, 89, 93, 86

99, 85, 87, 73, 90, 79, 89, 83

(i) Range

$$R = x_m - x_o$$

$$= 99 - 71$$

$$R = 28$$

(ii) K :-

$$K = 1 + 3.3 \log N$$

$$= 1 + 3.3 \log (35)$$

$$= 1 + 4.95$$

$$K = 5.95$$

(iii) h :-

$$h = R/K$$

$$= 28/5.95$$

$$= 4.70$$

$$h = 5$$

Teacher Feedback \_\_\_\_\_

Q No: 18

Decimal to Octal: $(932)_{10}$ 

$$(932)_{10} = (1500)_8$$

|   |       |
|---|-------|
| 8 | 932   |
| 8 | 104-0 |
| 8 | 13-0  |
|   | 15    |

Exercise

Q. No: 1 :-

Find frequency, cumulative frequency, Relative frequency class boundary and tellegam.

81, 86, 82, 76, 92, 89, 87, 82, 88

83, 85, 91, 77, 93, 83, 98, 92, 87

87, 82, 71, 88, 78, 84, 89, 93, 86

99, 85, 87, 73, 90, 79, 89, 83

(i) Range

$$R = x_m - x_o$$

$$= 99 - 71$$

$$R = 28$$

(ii) K :-

$$K = 1 + 3.3 \log N$$

$$= 1 + 3.3 \log (35)$$

$$= 1 + 4.95$$

$$K = 5.95$$

(iii) h :-

$$h = R/K$$

$$= 28/5.95$$

$$= 4.70$$

$$h = 5$$

Teacher Feedback \_\_\_\_\_



Date: .....

Day: .....

Q No: 16

Octal number to Binary

$$\begin{aligned}
 & (107)_8 \\
 &= 1 \times 8^2 + 0 \times 8^1 + 7 \times 8^0 \\
 &= 64 + 0 + 7 \\
 &= (71)_{10}
 \end{aligned}$$

$$\begin{array}{r|l}
 2 & 71 \\
 \hline
 2 & 35-1 \\
 \hline
 2 & 17-1 \\
 \hline
 2 & 8-1 \\
 \hline
 2 & 4-0 \\
 \hline
 2 & 2-0 \\
 \hline
 & 1-0
 \end{array}$$

$$(107)_8 = (1000111)_2$$

Q. No: 17

Octal equivalent

$$\begin{aligned}
 & (359.325)_{10} \\
 & 0.325 \times 8 = 2.5 \quad 2 \\
 & 0.5 \times 8 = 4.0 \quad 4 \\
 & 0.0 \times 8 = 0.0 \quad 0
 \end{aligned}$$

$$\begin{array}{r|l}
 8 & 359 \\
 \hline
 8 & 44-5 \\
 \hline
 & 5-4
 \end{array}$$

$$(359.325)_{10} = (545.246)_8$$

Marks: ..... Percent ..... % Grade .....

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$$(10110111)_2 = (255)_{10}$$

$$\begin{array}{r} 8 \overline{) 103} \\ 8 \overline{) 31-5} \\ \hline 7-5 \end{array}$$

Q No: 14

Binary to Octal number:

$$(110110.011)_2$$

$$= 1 \times 2^5 + 1 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 0 \times 2^0 + 0 \times 2^{-1} + 1 \times 2^{-2} + 1 \times 2^{-3}$$

$$= 32 + 16 + 0 + 4 + 2 + 0 + 0 + \frac{1}{4} + \frac{1}{8}$$

$$= 54 + 0.25 + 0.125$$

$$= (54.375)_{10}$$

$$0.375 \times 8 = 3.0$$

$$= 3.0$$

$$3$$

$$\begin{array}{r} 8 \overline{) 54} \\ 6-6 \end{array}$$

$$(110110.011)_2 = (66.3)_8$$

Q No: 15

Octal to Binary:

$$= (103.45)_8$$

$$= 1 \times 8^2 + 0 \times 8^1 + 3 \times 8^0 + 4 \times 8^{-1} + 5 \times 8^{-2}$$

$$= 64 + 0 + 3 + \frac{4}{8} + \frac{5}{64}$$

$$= 67 + 0.5 + 0.078125$$

$$(67.58)_{10}$$

$$0.58 \times 2 = 1.16$$

$$1.16$$

$$1$$

$$0.16 \times 2 = 0.32$$

$$0$$

$$\begin{array}{r} 2 \overline{) 67} \\ 2 \overline{) 33-1} \\ 2 \overline{) 16-1} \\ 2 \overline{) 8-0} \\ 1 \overline{) 4-0} \\ 2 \overline{) 2-0} \\ 1 \overline{) 0} \end{array}$$

$$(103.45)_8 = (1000011.10)_2$$

Teacher Feedback \_\_\_\_\_



Q No: 70

## Subtract the Binary

Number:

$$\begin{array}{r} (1100)_2 \\ - (1001)_2 \\ \hline (0011)_2 \end{array}$$

Q No: 12

## Binary to Octal

$$\begin{aligned} & (101101)_2 \\ &= 1 \times 2^5 + 0 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 \\ &= 32 + 0 + 8 + 4 + 0 + 1 \\ &= (45)_{10} \end{aligned}$$

$$(101101)_2 = (55)_8$$

$$\begin{array}{r} 8 \overline{) 45} \\ \underline{5-5} \\ 0 \end{array}$$

Q No: 13

## Binary to Octal

$$\begin{aligned} & (10110111)_2 \\ &= 1 \times 2^7 + 0 \times 2^6 + 1 \times 2^5 + 1 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 \\ &= 128 + 0 + 32 + 16 + 0 + 4 + 2 + 1 \\ &= (183)_{10} \end{aligned}$$

Marks ..... Percent ..... % Grade .....

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Q no: 7

Binary Addition :

$$\begin{array}{r} (1 \ 0 \ 1 \ 1)_2 \\ + (1 \ 1 \ 0)_2 \\ \hline (1 \ 0 \ 0 \ 0 \ 1)_2 \end{array}$$

Q no: 8

Binary Addition :

$$\begin{array}{r} (0 \ 1 \ 0 \ 0 \ 1 \ 1 \ 1)_2 \\ + (1 \ 1 \ 0 \ 1 \ 1 \ 0 \ 1)_2 \\ \hline (1 \ 0 \ 0 \ 1 \ 0 \ 1 \ 0 \ 0)_2 \end{array}$$

Q no: 9

Multiply of Binary Numbers:

$$\begin{array}{r} (1 \ 1 \ 1)_2 \\ \times (1 \ 0 \ 1)_2 \\ \hline 1 \ 1 \ 1 \\ 0 \ 0 \ 0 \ x \\ 1 \ 1 \ 1 \ x \ x \\ \hline (1 \ 0 \ 0 \ 0 \ 1 \ 1)_2 \end{array}$$

Teacher Feedback .....



Date: .....

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number: 0, 1, 2, 3, ..., 9

Q No: 4 :-

Binary to Decimal no.

$$\begin{aligned}
 & (10101)_2 \\
 &= 1 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 \\
 &= 16 + 0 + 4 + 0 + 1 \\
 &= (21)_{10}
 \end{aligned}$$

Q No: 5 :-

Binary to Decimal :

$$\begin{aligned}
 & (11111)_2 \\
 &= 1 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 \\
 &= 16 + 8 + 4 + 2 + 1 \\
 &= (29)_{10}
 \end{aligned}$$

Q No: 6 :-

Binary to Decimal

$$\begin{aligned}
 & (110011.11)_2 \\
 &= 1 \times 2^5 + 1 \times 2^4 + 0 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 + 1 \times 2^{-1} + 1 \times 2^{-2} \\
 &= 32 + 16 + 0 + 0 + 2 + 1 + \frac{1}{2} + \frac{1}{4} \\
 &= 51 + 0.5 + 0.25 \\
 &= (51.75)_{10}
 \end{aligned}$$

Marks ..... Percent ..... % Grade .....

Q no: 2 :-

### Octal Number :

The Octal number system is used extensively in digital work because it is easy to convert from octal to binary, vice versa.

The octal system has a base or radix of 8 which means that there are 8 symbols which are used to form octal number.

Q no: 3

### Decimal Number :

The decimal number system is a number of a base or radix equal to 10, which means that there are 10, called Arabic number symbol used to represent

Teacher Feedback: \_\_\_\_\_



Q. No: - 7

$18 \times 24$  to binary numbers  
 $= 18 \times 24$   
 $= (432)_{10}$

$$18 \times 24 = (110110000)_2$$

$$\begin{array}{r|l} 2 & 432 \\ \hline 2 & 216 - 0 \\ 2 & 108 - 0 \\ 2 & 54 - 0 \\ 2 & 27 - 0 \\ 2 & 13 - 1 \\ 2 & 6 - 1 \\ 2 & 3 - 0 \\ 2 & 1 - 1 \end{array}$$

## Short Questions

Q. No: 1 :

Define Binary number:

Binary number system is the number system of base or radix equal to 2, which means that there are two symbols used to represent number 0 and 1.

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$$\begin{array}{r}
 (c) \quad 1 \quad 0 \quad 1 \quad 0 \quad 1 \quad 1 \\
 - \quad 1 \quad 0 \quad 0 \quad 1 \quad 0 \quad 1 \\
 \hline
 0 \quad 0 \quad 0 \quad 1 \quad 1 \quad 0
 \end{array}$$

$$\begin{array}{r}
 (d) \quad 1 \quad 1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 1 \\
 - \quad 1 \quad 0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \\
 \hline
 0 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0
 \end{array}$$

Q no: 12 :-

Multiply the following.

$$\begin{array}{r}
 (a) \quad 1 \quad 1 \\
 \times 1 \quad 1 \\
 \hline
 1 \quad 1 \\
 1 \quad 1 \quad x \\
 \hline
 1 \quad 0 \quad 0 \quad 1
 \end{array}$$

$$\begin{array}{r}
 (b) \quad 1 \quad 0 \quad 1 \\
 \quad \quad 1 \quad 0 \\
 \quad \quad 0 \quad 0 \quad 0 \\
 \quad \quad 1 \quad 0 \quad 1 \quad x \\
 \hline
 1 \quad 0 \quad 1 \quad 0
 \end{array}$$

$$\begin{array}{r}
 (c) \quad 1 \quad 1 \quad 0 \\
 \times 1 \quad 0 \quad 1 \\
 \hline
 1 \quad 1 \quad 0 \\
 0 \quad 0 \quad 0 \quad x \\
 1 \quad 1 \quad 0 \quad x \quad x \\
 \hline
 1 \quad 1 \quad 1 \quad 1 \quad 0
 \end{array}$$

$$\begin{array}{r}
 (d) \quad 1 \quad 0 \quad 1 \quad 0 \\
 \times 1 \quad 0 \quad 1 \\
 \hline
 1 \quad 1 \quad 1 \quad 0 \\
 0 \quad 0 \quad 0 \quad 0 \quad x \\
 1 \quad 0 \quad 1 \quad 0 \quad x \quad x \\
 \hline
 1 \quad 1 \quad 0 \quad 1 \quad 1 \quad 0
 \end{array}$$

$$(e) \quad 1 \quad 0 \quad 1 \quad 1$$

$$\times 1 \quad 1 \quad 0 \quad 1$$

$$\hline
 1 \quad 0 \quad 1 \quad 1$$

$$0 \quad 0 \quad 0 \quad 0 \quad x$$

$$1 \quad 0 \quad 1 \quad 1 \quad x \quad x$$

$$1 \quad 0 \quad 1 \quad 1 \quad x \quad x$$

$$1 \quad 0 \quad 0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 1$$

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$$\begin{array}{r}
 00000 \\
 (c) \ 101011 \\
 110101 \\
 \hline
 1100000
 \end{array}$$

$$\begin{array}{r}
 000000 \\
 (d) \ 1110101 \\
 1011111 \\
 \hline
 11010100
 \end{array}$$

Q No: 10

Addition of Binary Numbers:

$$\begin{array}{r}
 0001 \\
 (a) \ 1101 \\
 1011 \\
 \hline
 100011
 \end{array}$$

$$\begin{array}{r}
 00000110 \\
 1110101 \\
 1001010 \\
 \hline
 100010101
 \end{array}$$

$$\begin{array}{r}
 000000 \\
 (c) \ 1111011 \\
 1011111 \\
 \hline
 100110000
 \end{array}$$

$$\begin{array}{r}
 000000 \\
 11010101 \\
 11010110 \\
 \hline
 1001100001
 \end{array}$$

Q No: 11

Subtract following binary numbers.

$$\begin{array}{r}
 11 \\
 (a) \ 10 \\
 \hline
 01
 \end{array}$$

$$\begin{array}{r}
 111 \\
 (b) \ 101 \\
 \hline
 010
 \end{array}$$

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$$(d) \ 10111.101$$

$$= 1 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 + 1 \times 2^{-1} + 0 \times 2^{-2} + 1 \times 2^{-3}$$

$$= 16 + 0 + 4 + 2 + 1 + \frac{1}{2} + 0 + \frac{1}{8}$$

$$= 23 + 0.5 + 0.125$$

$$(23.625)_{10}$$

$$0.625 \times 8 = 5$$

$$8 \overline{) 23}$$

$$\underline{2-7}$$

$$(10111.101)_2 = (27.5)_8$$

$$(e) \ 111010.001$$

$$= 1 \times 2^5 + 1 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 0 \times 2^0 + 0 \times 2^{-1} + 0 \times 2^{-2} + 1 \times 2^{-3}$$

$$= 32 + 16 + 8 + 0 + 2 + 0 + 0 + 0 + \frac{1}{8}$$

$$= 58 + 0.125$$

$$= (58.125)_{10}$$

$$0.125 \times 8 = 1$$

$$8 \overline{) 58}$$

$$\underline{7-2}$$

$$111010.001 = (72.1)_8$$

Q no: 9

Addition of Binary number:

a)

$$\begin{array}{r} \phantom{0}1 \phantom{0}1 \\ + \phantom{0}1 \phantom{0}1 \\ \hline 1 \phantom{0}1 \phantom{0}0 \end{array}$$

(b)

$$\begin{array}{r} \phantom{0}0 \phantom{0}0 \phantom{0}0 \\ \phantom{0}1 \phantom{0}0 \phantom{0}1 \phantom{0}1 \\ + \phantom{0}1 \phantom{0}1 \phantom{0}0 \phantom{0}1 \\ \hline 1 \phantom{0}1 \phantom{0}0 \phantom{0}0 \phantom{0}0 \end{array}$$

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$$\begin{aligned}
 (c) \quad 65_{10} &:= \\
 &= 6 \times 10^1 + 5 \times 10^0 \\
 &= 40 + 5 \\
 &= (53)_{10}
 \end{aligned}$$

$$(65)_{10} = (110101)_2$$

$$\begin{array}{r|l}
 2 & 53 \\
 \hline
 2 & 26-1 \\
 2 & 13-0 \\
 2 & 6-1 \\
 2 & 3-0 \\
 & 1-1
 \end{array}$$

$$(d) \quad 124.375 :$$

$$= 1 \times 10^2 + 2 \times 10^1 + 4 \times 10^0 + 3 \times 10^{-1} + 7 \times 10^{-2} + 5 \times 10^{-3}$$

$$= 64 + 16 + 4 + 3/10 + 7/100 + 5/1000$$

$$= 84 + 0.375 + 0.109375 + 0.009765625$$

$$= (84.494)_{10}$$

$$0.494 \times 2 = 0.988 \quad 0$$

$$0.988 \times 2 = 1.976 \quad 1$$

$$0.976 \times 2 = 1.952 \quad 1$$

$$0.952 \times 2 = 1.904 \quad 1$$

$$0.904 \times 2 = 1.808 \quad 1$$

$$0.808 \times 2 = 1.616 \quad 1$$

$$0.616 \times 2 = 1.232 \quad 1$$

$$0.232 \times 2 = 0.464 \quad 0$$

$$0.464 \times 2 = 0.928 \quad 0$$

$$0.928 \times 2 = 1.856 \quad 1$$

$$(124.375)_{10} = (1010100.011111001)_2$$

نوٹ: ہر عدد "values" ہو گی انہی بار

E.g:- 84.494

"Multiply" کرنا ہے۔

صرف 1 بار "Multiply" کرنا ہے

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(h)  $49.21875$  :

$$0.21875 \times 2 = 1.75$$

$$0.75 \times 2 = 6.0$$

$$\begin{array}{r|l} 2 & 49 \\ \hline & 6-1 \end{array}$$

$$(49.21875)_{10} = (61.6)_2$$

Q No: 5

Convert Octal number to their Binary :

(a)  $13_8$  :

$$= 1 \times 8^1 + 3 \times 8^0$$

$$= 8 + 3$$

$$= (11)_{10}$$

$$\begin{array}{r|l} 2 & 11 \\ \hline 2 & 5-1 \\ 2 & 2-1 \\ \hline & 1-0 \end{array}$$

$$(13)_8 = (1011)_2$$

(b)  $27_8$  :

$$= 2 \times 8^1 + 7 \times 8^0$$

$$= 16 + 7$$

$$= (23)_{10}$$

$$\begin{array}{r|l} 2 & 23 \\ \hline 2 & 11-1 \\ 2 & 5-1 \\ 2 & 2-1 \\ \hline & 1-0 \end{array}$$

$$(27)_8 = (10111)_2$$

Marks \_\_\_\_\_ Percent \_\_\_\_\_ % Grade \_\_\_\_\_



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(c) 125 :

$$(125)_{10} = (175)_8$$

$$\begin{array}{r|l} 8 & 125 \\ 8 & 15-5 \\ \hline & 1-7 \end{array}$$

(d) 243.67 :

$$0.67 \times 8 = 5.36$$

5

$$0.36 \times 8 = 2.88$$

2

$$0.88 \times 8 = 7.04$$

7

$$\begin{array}{r|l} 8 & 243 \\ 8 & 30-3 \\ \hline & 3-6 \end{array}$$

$$(243.67)_{10} = (363.527)_8$$

(e) 419.95 :

$$0.95 \times 8 = 7.6$$

7

$$0.6 \times 8 = 4.8$$

4

$$0.8 \times 8 = 6.4$$

6

$$\begin{array}{r|l} 8 & 419 \\ 8 & 52-3 \\ \hline & 6-4 \end{array}$$

$$(419.95)_{10} = (643.746)_8$$

(f) 645.7 :

$$0.07 \times 8 = 0.56$$

0

$$0.56 \times 8 = 4.48$$

4

$$0.48 \times 8 = 3.84$$

3

$$\begin{array}{r|l} 8 & 645 \\ 8 & 80-5 \\ 8 & 10-0 \\ \hline & 1-2 \end{array}$$

$$(645.7)_{10} = (1205.043)_8$$

(g) 39.4475 :-

$$0.4475 \times 8 = 3.58$$

3

$$0.58 \times 8 = 4.64$$

4

$$0.64 \times 8 = 5.12$$

5

$$\begin{array}{r|l} 8 & 39 \\ & 4-7 \end{array}$$

$$(39.4475)_{10} = (47.345)_8$$

Teacher Feedback \_\_\_\_\_

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$$= 0.25 + 0.078125 + 0.0078125$$

$$= (0.3359)_{10}$$

$$(7) \quad 16742.3_8 :$$

$$= 1 \times 8^4 + 6 \times 8^3 + 7 \times 8^2 + 4 \times 8^1 + 2 \times 8^0 + 3 \times 8^{-1}$$

$$= 4096 + 3072 + 448 + 32 + 2 + \frac{3}{8}$$

$$= 7650 + 0.375$$

$$= 7650.375$$

$$(9) \quad 206.104_8 :$$

$$= 2 \times 8^2 + 0 \times 8^1 + 6 \times 8^0 + 1 \times 8^{-1} + 0 \times 8^{-2} + 4 \times 8^{-3}$$

$$= 128 + 0 + 6 + \frac{1}{8} + 0 + \frac{4}{512}$$

$$= 128 + 6 + 0.125 + 0.0078125$$

$$= (134.1328)_{10}$$

Q No 4 :-

Decimal to Octal :

$$(a) \quad 29 :$$

$$\begin{array}{r|l} 8 & 29 \\ & 3-5 \end{array}$$

$$(29)_{10} = (35)_8$$

$$(b) \quad 68 :$$

$$\begin{array}{r|l} 8 & 68 \\ 8 & 8-4 \\ & 1-0 \end{array}$$

$$(68)_{10} = (104)_8$$

Marks ..... Percent ..... % Grade .....



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 $(674)_{10} : (110101110)_2$ 

Q No: - 3

Octal to Decimal number:

a)  $14_8 :$ 

$$= 1 \times 8^1 + 4 \times 8^0$$

$$= 8 + 4$$

$$= (12)_{10}$$

(b)  $236_8 =$ 

$$= 2 \times 8^2 + 3 \times 8^1 + 6 \times 8^0$$

$$= 128 + 24 + 6$$

$$= (158)_{10}$$

(c)  $1432_8 :$ 

$$= 1 \times 8^3 + 4 \times 8^2 + 3 \times 8^1 + 2 \times 8^0$$

$$= 512 + 256 + 24 + 2$$

$$(794)_{10}$$

(d)  $0.43_8 :$ 

$$= 0 \times 8^0 + 4 \times 8^{-1} + 3 \times 8^{-2}$$

$$= 0 + \frac{4}{8} + \frac{3}{64}$$

$$= 0.5 + 0.046875$$

$$= (0.5469)_{10}$$

(e)  $0.254_8 :$ 

$$= 0 \times 8^0 + 2 \times 8^{-1} + 5 \times 8^{-2} + 4 \times 8^{-3}$$

$$= 0 + \frac{2}{8} + \frac{5}{64} + \frac{4}{512}$$

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$$0.76 \times 2 = 1.52 \quad 1$$

$$0.52 \times 2 = 1.04 \quad 1$$

$$(962.84)_{10} = (1111000010.11010111)_2$$

(d) 0.0132 :

$$0.0132 \times 2 = 0.0264 \quad 0$$

$$0.0264 \times 2 = 0.0528 \quad 0$$

$$0.0528 \times 2 = 0.1056 \quad 0$$

$$0.1056 \times 2 = 0.2112 \quad 0$$

$$0.2112 \times 2 = 0.4224 \quad 0$$

$$0.4224 \times 2 = 0.8448 \quad 0$$

$$0.8448 \times 2 = 1.6896 \quad 1$$

$$0.6896 \times 2 = 1.3792 \quad 1$$

$$0.3792 \times 2 = 0.7584 \quad 0$$

$$0.7584 \times 2 = 1.5168 \quad 1$$

$$0.5168 \times 2 = 1.0336 \quad 1$$

$$0.0336 \times 2 = 0.672 \quad 0$$

$$(0.0132)_{10} = (0.000000110110)_2$$

(e) 6.74 :

$$0.74 \times 2 = 1.48 \quad 1$$

$$0.48 \times 2 = 0.96 \quad 0$$

$$0.96 \times 2 = 1.92 \quad 1$$

$$0.92 \times 2 = 1.84 \quad 1$$

$$0.84 \times 2 = 1.68 \quad 1$$

$$0.68 \times 2 = 1.36 \quad 1$$

$$0.36 \times 2 = 0.72 \quad 0$$

$$\begin{array}{r|l} 2 & 6 \\ \hline 2 & 3-0 \\ & 1-1 \end{array}$$

Marks \_\_\_\_\_ Percent \_\_\_\_\_ % Grade \_\_\_\_\_



Date: .....

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Q No: 2 :

Decimal to Binary number:

(a) 16 :

|   |     |
|---|-----|
| 2 | 16  |
| 2 | 8-0 |
|   | 4-0 |
| 2 | 2-0 |
|   | 1-0 |

$$(16)_{10} = (10000)_2$$

(b) 247 :

|   |       |
|---|-------|
| 2 | 247   |
| 2 | 123-1 |
| 2 | 61-1  |
| 2 | 30-1  |
| 2 | 15-0  |
| 2 | 7-1   |
| 2 | 3-1   |
|   | 1-1   |

$$(247)_{10} = (11110111)_2$$

(c) 962.84 :

$$0.84 \times 2 = 1.68$$

↓

$$2 \mid 962$$

$$0.68 \times 2 = 1.36$$

1

$$2 \mid 481-0$$

$$0.36 \times 2 = 0.72$$

0

$$2 \mid 240-1$$

$$0.72 \times 2 = 1.44$$

1

$$2 \mid 120-0$$

$$0.44 \times 2 = 0.88$$

0

$$2 \mid 60-0$$

$$0.88 \times 2 = 1.76$$

1

$$2 \mid 30-0$$

$$2 \mid 15-0$$

$$2 \mid 7-1$$

Teacher Feedback

$$= (1.125)_{10}$$

$$(e) \quad (110100.010011)_2 :$$

$$= 1 \times 2^5 + 1 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 0 \times 2^0 + 0 \times 2^{-1} + 1 \times 2^{-2} + 0 \times 2^{-3} \\ + 0 \times 2^{-4} + 1 \times 2^{-5} + 1 \times 2^{-6}$$

$$= 32 + 16 + 0 + 4 + 0 + 0 + 0 + \frac{1}{4} + 0 + 0 + \frac{1}{32} + \frac{1}{64}$$

$$= 32 + 20 + 0.25 + 0.03125 + 0.0156$$

$$= (59.2969)_{10}$$

$$(f) \quad (11010.10110)_2 :$$

$$= 1 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 0 \times 2^0 + 1 \times 2^{-1} + 0 \times 2^{-2} + 1 \times 2^{-3} \\ + 1 \times 2^{-4} + 0 \times 2^{-5}$$

$$= 16 + 8 + 0 + 2 + 0 + \frac{1}{2} + 0 + \frac{1}{8} + \frac{1}{32} + 0$$

$$= 26 + 0.5 + 0.125 + 0.03125$$

$$= (26.656)_{10}$$

$$(g) \quad (1000001.111)_2 :$$

$$= 1 \times 2^6 + 0 \times 2^5 + 0 \times 2^4 + 0 \times 2^3 + 0 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 + \\ 1 \times 2^{-1} + 1 \times 2^{-2} + 1 \times 2^{-3}$$

$$= 64 + 0 + 0 + 0 + 0 + 0 + 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8}$$

$$= 65 + 0.5 + 0.25 + 0.125$$

$$= (65.875)_{10}$$



$$= (1.125)_{10}$$

$$(e) (110100.010011)_2 :$$

$$= 1 \times 2^5 + 1 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 0 \times 2^0 + 0 \times 2^{-1} + 1 \times 2^{-2} + 0 \times 2^{-3} \\ + 0 \times 2^{-4} + 1 \times 2^{-5} + 1 \times 2^{-6}$$

$$= 32 + 16 + 0 + 4 + 0 + 0 + 0 + \frac{1}{4} + 0 + 0 + \frac{1}{32} + \frac{1}{64}$$

$$= 32 + 20 + 0.25 + 0.03125 + 0.0156$$

$$= (59.2969)_{10}$$

$$(f) (11010.10110)_2 :$$

$$= 1 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 0 \times 2^0 + 1 \times 2^{-1} + 0 \times 2^{-2} + 1 \times 2^{-3} \\ + 1 \times 2^{-4} + 0 \times 2^{-5}$$

$$= 16 + 8 + 0 + 2 + 0 + \frac{1}{2} + 0 + \frac{1}{8} + \frac{1}{32} + 0$$

$$= 26 + 0.5 + 0.125 + 0.03125$$

$$= (26.656)_{10}$$

$$(g) (1000001.111)_2 :$$

$$= 1 \times 2^6 + 0 \times 2^5 + 0 \times 2^4 + 0 \times 2^3 + 0 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 + \\ 1 \times 2^{-1} + 1 \times 2^{-2} + 1 \times 2^{-3}$$

$$= 64 + 0 + 0 + 0 + 0 + 0 + 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8}$$

$$= 65 + 0.5 + 0.25 + 0.125$$

$$= (65.875)_{10}$$

Q<sub>no</sub>: 1Binary to Decimal number.(a)  $(100)_2$  :

$$= 1 \times 2^2 + 0 \times 2^1 + 0 \times 2^0$$

$$= 4 + 0 + 0$$

$$= (4)_{10}$$

(b)  $(11010)_2$  :

$$= 1 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 0 \times 2^0$$

$$= 16 + 8 + 0 + 2 + 0$$

$$= 24 + 2$$

$$= (26)_{10}$$

(c)  $(10110010)_2$  :

$$= 1 \times 2^7 + 0 \times 2^6 + 1 \times 2^5 + 1 \times 2^4 + 0 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 0 \times 2^0$$

$$= 128 + 0 + 32 + 16 + 0 + 0 + 2 + 0$$

$$= 128 + 50$$

$$= (178)_{10}$$

(d)  $(1.001)_2$  :

$$= 1 \times 2^0 + 0 \times 2^{-1} + 0 \times 2^{-2} + 1 \times 2^{-3}$$

$$= 1 + 0 + 0 + \frac{1}{8}$$

$$= 1 + 0.125$$

Teacher Feedback: \_\_\_\_\_